

STOLNICI, Romania

WMO: 154160

Lat: 44.5628N

Lon: 24.7897E

Elev: 210

StdP: 98.83

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.1	-13.9	1.2	-12.0	-11.4	1.4	-9.0	13.3	0.9	10.3	0.8	0.8	330	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.4	33.3	20.6	31.9	20.3	30.4	20.0	22.6	30.0	21.8	28.8	21.1	27.9	2.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.4	25.0	19.5	14.6	24.4	18.8	14.0	23.7	67.3	30.0	64.5	29.0	61.9	28.1	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	7.4	6.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-1.9	2.0	5.9	11.1	16.6	21.1	22.3	22.9	17.6	11.5	5.3	1.1
	DBStd	9.28	4.79	4.71	4.48	4.25	3.05	2.78	2.67	2.94	3.89	3.63	4.35	4.26
	HDD10.0	1224	369	224	137	38	1	0	0	1	27	151	277	
	HDD18.3	2969	627	456	385	218	68	7	3	61	213	392	536	
	CDD10.0	1712	0	1	10	70	207	333	382	400	228	73	9	0
	CDD18.3	418	0	0	0	1	16	91	127	146	38	1	0	0
	CDH23.3	4404	0	0	1	36	188	887	1294	1561	411	27	0	0
	CDH26.7	1657	0	0	0	2	27	304	493	701	128	1	0	0
(14) Wind	WSAvg	2.2	2.0	2.4	2.4	2.8	2.4	2.0	1.8	2.1	2.2	2.1	1.9	1.9
(15) Precipitation	PrecAvg	620	37	29	39	57	71	79	77	45	57	50	38	43
	PrecMax	1088	76	68	102	171	143	148	173	111	176	126	101	147
	PrecMin	299	6	2	2	19	12	22	17	7	2	0	0	0
	PrecStd	162	22	18	27	33	33	36	41	27	44	31	26	34
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	17.2	22.1	26.3	29.3	34.3	34.2	36.3	32.4	25.9	19.8	14.0
		MCWB	7.6	11.1	13.3	16.2	19.3	21.6	21.7	21.5	19.3	15.9	13.4	9.3
	2%	DB	8.1	13.6	19.0	24.1	27.0	31.7	32.7	33.7	30.0	23.5	16.3	11.0
		MCWB	5.8	9.4	11.6	15.5	18.1	20.9	21.2	20.1	18.2	15.3	12.2	8.0
	5%	DB	5.9	11.0	16.3	21.5	25.4	29.9	31.2	32.3	27.7	20.9	13.4	8.6
		MCWB	4.4	7.8	10.4	14.2	17.4	20.5	20.8	20.0	17.5	14.1	10.8	6.5
	10%	DB	4.1	8.7	13.7	19.2	23.6	28.3	29.6	30.6	25.3	18.6	11.3	6.5
		MCWB	3.1	6.3	8.8	13.0	16.5	19.9	20.4	19.7	16.9	13.4	9.6	5.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	12.1	14.0	17.1	20.2	23.5	23.6	23.5	20.7	16.8	14.2	9.8
		MCDB	11.1	16.3	21.2	24.6	27.6	31.2	30.7	31.7	28.8	22.7	18.4	11.9
	2%	WB	6.3	9.7	12.3	16.1	19.1	22.2	22.5	22.2	19.3	15.8	12.9	8.4
		MCDB	7.8	13.1	18.2	23.6	25.5	29.1	30.0	29.9	26.1	22.1	15.9	11.0
	5%	WB	4.8	8.2	10.7	14.9	18.1	21.4	21.9	21.4	18.4	14.9	11.4	6.9
		MCDB	6.1	10.7	15.7	21.0	24.0	28.0	28.8	28.8	25.1	20.0	13.2	8.8
	10%	WB	3.3	6.6	9.2	13.7	17.1	20.5	21.1	20.6	17.6	13.7	10.1	5.4
		MCDB	4.1	8.3	13.2	18.2	22.7	27.0	27.4	28.1	24.3	17.7	11.6	6.6
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.2	9.8	11.1	11.3	12.0	12.6	13.4	12.0	10.7	7.4	7.1
		MCDBR	9.7	13.0	15.5	15.5	14.2	14.6	14.8	15.8	15.7	15.8	11.8	10.6
	5% WB	MCWBR	7.0	9.0	8.9	8.1	6.3	5.5	4.9	4.5	5.6	8.2	7.3	7.8
		MCWBR	9.0	12.6	14.0	13.9	13.3	13.3	12.9	13.4	13.1	14.1	10.3	10.9
(40) Clear-Sky Solar Irradiance	taub		0.332	0.363	0.399	0.457	0.434	0.439	0.450	0.469	0.427	0.389	0.364	0.336
		taud	2.309	2.235	2.196	2.049	2.189	2.210	2.186	2.127	2.208	2.306	2.323	2.331
	Ebn at Noon		767	805	826	805	840	834	819	786	787	768	721	721
		Edh at Noon	82	106	126	159	142	140	142	145	124	98	82	74
(44) All-Sky Solar Radiation	RadAvg		1.42	2.26	3.48	4.72	5.82	6.39	6.45	5.68	4.12	2.69	1.50	1.15
	RadStd		0.21	0.35	0.39	0.46	0.44	0.45	0.42	0.37	0.33	0.34	0.19	0.17

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (71 neighbors)		+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-137	+60	N/A	

Nomenclature: See separate page